FOR REPRESENTABLE MORPHISMS THE VALUATIVE CRITERION OF ALGEBRAIC SPACES IS THE VALUATIVE CRITERION OF SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof uses only the pushing-down trick and the valuation-ring property K intersect A' = A; the representability hypothesis is what makes X_A a scheme and hence the local ring argument available.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The valuative criterion as formulated for algebraic spaces. Definition definition-valuative-criterion, used at proof lines 8392-8393 and 8409-8417. Fixes the K'/A' diagrams that the proof pushes down.
- The valuative criterion as formulated for schemes. Schemes, Definition schemes-definition-valuative-criterion, used at proof lines 8401-8404. Is the second of the two equivalent conditions.
- Representability of f makes X_A = Spec(A) x_Y X a scheme. used at proof lines 8418-8420. Lets the point x and its local ring be used.
- Local rings and points of schemes as ring maps into fields. Schemes, Section schemes-section-points, used at proof lines 8429-8432. Converts the ring map O_{X_A,x} -> A into a morphism Spec(A) -> X_A.

Source reference. Original definition 03IX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4619–4619. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 01KD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2046–2046. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 01J5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2002–2002. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is representable. The following are equivalent*

(1) *f satisfies the existence part of the valuative criterion as in Definition 03IX,*
(2) *given any commutative solid diagram*

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

where A is a valuation ring with field of fractions K, there exists a dotted arrow, i.e., f satisfies the existence part of the valuative criterion as in Schemes, Definition 01KD.

Proof. It suffices to show that given a commutative diagram of the form

$$\begin{array}{ccccc} \mathrm{Spec}(K') & \longrightarrow & \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & & & \overset{\varphi}{\nearrow} & \downarrow \\ \mathrm{Spec}(A') & \longrightarrow & \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

as in Definition 03IX, then we can find a morphism $\mathrm{Spec}(A) \to X$ fitting into the diagram too. Set $X_A = \mathrm{Spec}(A) \times_Y Y$. As f is representable we see that X_A is a scheme. The morphism φ gives a morphism $\varphi' : \mathrm{Spec}(A') \to X_A$. Let $x \in X_A$ be the image of the closed point of $\varphi' : \mathrm{Spec}(A') \to X_A$. Then we have the following commutative diagram of rings

$$\begin{array}{ccccc} K' & \longleftarrow & K & \longleftarrow & \mathcal{O}_{X_A,x} \\ \uparrow & & & \swarrow & \uparrow \\ A' & \longleftarrow & A & \longleftarrow & A \end{array}$$

Since A is a valuation ring, and since A' dominates A, we see that $K \cap A' = A$. Hence the ring map $\mathcal{O}_{X_A,x} \to K$ has image contained in A. Whence a morphism $\mathrm{Spec}(A) \to X_A$ (see Schemes, Section 01J5) as desired. □